

Managing Disruptive Technologies

Assignment 5

Brendan Reed | breed | A4

Question 1

(a) Column (1):

The Treatment coefficient in column (1) is 0.0007, with a standard error of 0.00004, yielding a t-statistic of approximately 17.5. The result is statistically significant at conventional thresholds.

Magnitude: The coefficient of assigning the algorithmic recommendation increases clicks on slot 4 by 0.0007 per user-session, against a baseline of 0.0277 clicks per session. An increase of approximately: $0.0007 / 0.0277 \approx 2.5\%$.

Practical significance: 0.0007 additional clicks per session is small in absolute terms. Although, the platform serves over 120 million monthly page impressions (background doc). At this scale, a 2.5% engagement increase on a single slot produces significant advertising revenue. The effect is modestly practically significant but not transformative. It is commercially meaningful at platform scale.

Recommended strategy: The algorithm outperforms human curation in the full sample. A CTO using these results is justified in deploying algorithmic recommendations for slot 4, since the effect is statistically robust and relevant with no observed downside.

(b) Column (2): Cold-Start

$$TE(Y) = -0.0081 + 0.0030 \times \ln(Y)$$

Both coefficients are statistically significant ($p < 0.001$).

Finding breakeven point:

$$TE(Y) = 0: 0.0030 \times \ln(Y) = 0.0081$$

$\ln(Y) = 2.70$ and $Y \approx 14.9$ prior visits.

Users with fewer than approximately 15 prior visits experience a negative treatment effect. i.e. the algorithm harms engagement relative to the human editor for these low-frequency users.

Prior Visits (Y)	ln(Y)	TE(Y) = $-0.0081 + 0.0030 \times \ln(Y)$	Algorithm vs. Human
1	0.00	-0.0081	Human
5	1.61	-0.0033	Human
10	2.30	-0.0012	Human
15	2.71	≈ 0.0000	Breakeven
20	3.00	+0.0009	Algorithm
50	3.91	+0.0036	Algorithm
100	4.61	+0.0057	Algorithm

Interpretation: For users without prior visit history, the algorithm falls back on collaborative filtering based on similar users. This generic signal is weaker than experienced judgment, producing the cold-start penalty.

The positive interaction term (+0.0030) reflects data network effects; each additional visit generates personalized data, training the algorithm to shift from toward individual information filtering and outperforming the human curator.

Practical significance: The treatment effect spans from -0.0081 (new users) to +0.0057 (100-visit users). This range of nearly 20 times the column (1) average is statistically and practically significant.

Revised strategy:

- **Fewer than ~15 prior visits:** Retain human curation (cold-start, algorithm is harmful).
- **15 or more prior visits:** Deploy algorithm recommendations.

(c) Column (3): Diminishing Marginal Returns to Data

Full treatment effect expression becomes:

$$TE(Y) = -0.0125 + 0.0084 \times \ln(Y) - 0.0009 \times [\ln(Y)]^2$$

The negative coefficient of Treatment * Log(Prior)² means the additional units of prior-visits to algorithmic performance is decreasing. This is diminishing marginal returns to data where beyond a certain data threshold, additional behavior history adds progressively less value.

Breakeven points:

$$TE(Y) = 0 \quad L = \ln(Y): 0.0009L^2 - 0.0084L + 0.0125 = 0$$

$L_1 \approx 1.86 \rightarrow Y_1 \approx 6.4$ visits

$L_2 \approx 7.47 \rightarrow Y_2 \approx 1,756$ visits

Peak treatment effect: $Y^* \approx 106$ visits.

Point of maximum algorithmic advantage.

Prior Visits (Y)	ln(Y)	TE(Y)	Algorithm vs. Human
1	0.00	-0.0125	Human
6	1.79	≈ 0.0002	Breakeven
10	2.30	+0.0021	Algorithm
106 (peak)	4.67	+0.0071 (max)	Algorithm
500	6.21	+0.0047	Diminishing returns
1,000	6.91	+0.0026	Further
1,756 (upper breakeven)	7.47	≈ 0.0000	breakeven

Column (3) reveals a relationship where the algorithm outperforms the human editor within the window of roughly 7 to 1,756 visits. This performance peaks at approximately 106. For frequent visitors above this threshold 1,756, the human editor retains an advantage.

A possible reason for this is the model tries to over-personalize based on stale preferences, reducing content discovery.

Revised strategy: A CTO should implement a three-tier policy based on the observable *past_visits*:

- **Fewer than ~7 prior visits:** Human curation
- **7 to ~1,756 prior visits:** Algorithmic recommendation (algorithm outperforms; peak at ~106 visits).
- **More than ~1,756 prior visits:** Return to human curation or a hybrid approach (algorithm performance declines below human baseline due to diminishing and negative marginal returns to data).